

Theoretical Exercise 4

1) We will now calculate the log-likelihood:

$$\begin{aligned}
 \ell(w; D) &= \sum_{i=1}^n \log [p_{x|w}(x^{(i)}, y^{(i)})] = \sum_{i=1}^n \log [p_{x|w}(x^{(i)}) p_{y|x,w}(y^{(i)})] = \\
 &= \sum_{i=1}^n \log [p_{x|w}(x^{(i)})] + \sum_{i=1}^n \log [p_{y|x,w}(y^{(i)})] = \quad \text{↖ } p_{x|w}(x^{(i)}) \text{ does not depend on the weight vector } w. \\
 &= \sum_{i=1}^n \log [p_x(x^{(i)})] + \sum_{i=1}^n \log [p(y = y^{(i)} | x^{(i)}, w)] = \\
 &= \text{const}(w) + \sum_{i=1}^n \log [p(y = y^{(i)} | x^{(i)}, w)] \quad (*) \\
 &= \sum_{i=1}^n y^{(i)} \log [\phi(w^T x^{(i)})] + (1 - y^{(i)}) \log [1 - \phi(w^T x^{(i)})]
 \end{aligned}$$

$$\begin{aligned}
 (*) \log [p(y = y^{(i)} | x^{(i)}, w)] &= \begin{cases} \log [\phi(w^T x^{(i)})], & y^{(i)} = 1 \\ \log [1 - \phi(w^T x^{(i)})], & y^{(i)} = 0 \end{cases} = \\
 &= y^{(i)} \log [\phi(w^T x^{(i)})] + (1 - y^{(i)}) \log [1 - \phi(w^T x^{(i)})]
 \end{aligned}$$

2) We want to maximize the log-likelihood, meaning we wish to compute:

$$\arg \max_{w \in \mathbb{R}^{r+1}} \sum_{i=1}^n y^{(i)} \log [\phi(w^T x^{(i)})] + (1 - y^{(i)}) \log [1 - \phi(w^T x^{(i)})]$$

Since this expression is negative (prob. are less than 1), maximization is often cast as minimization:

(It was stated in Piazza to use $\log_e = \ln$)

$$\arg \min_{w \in \mathbb{R}^{p+1}} \sum_{i=1}^n -y^{(i)} \ln [\phi(w^T x^{(i)})] - (1 - y^{(i)}) \ln [1 - \phi(w^T x^{(i)})]$$

Thus, we get that the BCE loss can be represented like so:

$$L_{\text{BCE}}(w) = - \sum_{i=1}^n y^{(i)} \ln [\phi(w^T x^{(i)})] + (1 - y^{(i)}) \ln [1 - \phi(w^T x^{(i)})]$$

3) The gradient of the BCE loss is:

$$\nabla L_{\text{BCE}}(w) = \nabla \left(- \sum_{i=1}^n y^{(i)} \ln [\phi(w^T x^{(i)})] + (1 - y^{(i)}) \ln [1 - \phi(w^T x^{(i)})] \right) =$$

$$= - \sum_{i=1}^n y^{(i)} \cdot \frac{x^{(i)}}{\phi(w^T x^{(i)})} \cdot \phi(w^T x^{(i)}) + (1 - y^{(i)}) \cdot \frac{-x^{(i)}}{(1 - \phi(w^T x^{(i)}))} \cdot \phi(w^T x^{(i)}) =$$

where $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ is the derivative of $\Phi(x)$

$$= - \sum_{i=1}^n \left(\frac{y^{(i)}}{\phi(w^T x^{(i)})} - \frac{(1 - y^{(i)})}{(1 - \phi(w^T x^{(i)}))} \right) x^{(i)} \phi(w^T x^{(i)})$$

We minimize it using gradient descent:

* Initialize w randomly

Until convergence, do:

$$w_{t+1} = w_t - \eta \nabla L_{BCE}(w_t)$$

h) As the data already has the value 1 for the first feature in all instances, there is no need to add a bias term.

5)

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Iteration #1:
Gradient: [-0.31584773 -6.10999505 -3.37822655 -3.60269169 -0.37801795]
BCE Loss: 4.436048049671651

Iteration #2:
Gradient: [ 0.14536828 -5.1611566 -2.60612494 -2.79307027  0.80357949]
BCE Loss: 3.8772995168646127

Iteration #3:
Gradient: [ 0.33261427 -4.54851285 -2.08508581 -2.37888589  1.20653745]
BCE Loss: 3.485421250058467

Iteration #4:
Gradient: [ 0.41007571 -4.10160363 -1.70097696 -2.12886882  1.31369774]
BCE Loss: 3.179408600847891
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6)

Sample	True	Pred	Correct
1	1	1	Y
2	0	0	Y
3	1	1	Y
4	0	1	N
5	1	1	Y
6	0	1	N
7	1	1	Y
8	1	1	Y

Accuracy: 75.00%

Misclassified samples: [4, 6]